

Eulerian orientations of separable 4-regular graphs on sixteen vertices

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Abstract

Among simple connected separable 4-regular graphs on sixteen vertices, the maximum number of Eulerian orientations is 9216, attained by exactly one isomorphism class. We describe the extremizer explicitly and give an elementary count of its orientations. Since $(4 \cdot 9216)^7 / 48^{19} = 256/243 > 1$, this graph refutes Conjecture 1 of Bartzos and Samaris, arXiv:2609.06701v1. The matching upper bound and uniqueness are computer-assisted: a cut-vertex reduction leaves six branch orders, an elementary complete generator produces 3205 representatives with duplicates retained, and exact orientation counts determine all possible equality cases. Separate exhaustive and meet-in-the-middle computations reproduce every representative count. No extremal formula at other orders is asserted.

1 Statement and relation to the proposed bound

For a finite undirected graph or loopless multigraph Q with all degrees even, let $\text{EO}(Q)$ denote the number of orientations with equal indegree and outdegree at every vertex. Each edge occurrence receives its own direction, including parallel edges. We do not identify orientations under automorphisms or global reversal. A connected graph is *separable* if it has a cut vertex. Let $\mathcal{C}_{16}$ be the class of simple connected separable 4-regular graphs of order 16.

Bartzos and Samaris [1, Conjecture 1, p. 20] proposed that every simple connected separable 4-regular graph G of order n satisfies

$$\text{EO}(G) \leq \frac{1}{4} 48^{(n+3)/7}. \quad (1)$$

The conjecture in that version has no congruence restriction on n . Their exhaustive computations cover $n \leq 15$ [1, Table 1, pp. 15–16]. We determine the next finite maximum and its equality class.

Define W on the vertex set $\{0, 1, \dots, 15\}$ by the following three pieces, with no edges other than those listed:

$$\begin{aligned} L &= (K_{\{0,1,2,3,4\}} - \{(0, 1)\}) + \{(0, 14), (1, 14)\}, \\ C &= K_{\{10,11,12,13\}} + \{(10, 14), (11, 14), (12, 15), (13, 15)\}, \\ R &= (K_{\{5,6,7,8,9\}} - \{(5, 6)\}) + \{(5, 15), (6, 15)\}. \end{aligned}$$

Here K_S denotes the complete graph on the displayed vertex set, and each pair denotes an unordered edge. The pieces L and C meet only at vertex 14; C and R meet only at vertex 15.

Theorem 1. *Every $G \in \mathcal{C}_{16}$ satisfies $\text{EO}(G) \leq 9216$, with equality if and only if G is isomorphic to W . Furthermore,*

$$\text{EO}(W) = 9216 > \frac{1}{4} 48^{19/7}.$$

Consequently, inequality (1) is false at $n = 16$, and the universal conjecture as stated in the cited version is false.

The orientation count for W and the strict comparison are elementary. The upper bound and equality classification use the finite computation described below. Cut-flow factorization is already present in [1, Lemma 6 and Theorem 4]; we prove the precise special case needed here. The contribution relative to that version is the counterexample together with the exact order-16 maximum and complete equality classification. For adjacent context on exact finite reductions for 4-regular graphs, see Zeng [2]. That work concerns girth constraints; it is not used to establish any Eulerian-orientation count in this paper. Our source comparison is with arXiv:2609.06701v1, and we make no priority claim relative to later versions or other work.

2 An elementary counterexample

We first verify the lower bound and disproof independently of the census. The definition of W gives 32 distinct nonloop edges. Every vertex has degree four, the graph is connected, and deleting 14 separates $\{0, \dots, 4\}$ from the remaining vertices. Thus $W \in \mathcal{C}_{16}$.

Lemma 2. *On four fixed vertices, there are exactly four tournament orientations with any prescribed assignment of outdegrees $(2, 2, 1, 1)$. Consequently $\text{EO}(K_5) = 24$.*

Proof. Choose the direction of the edge between the two vertices whose required outdegrees are two. Its head must beat both vertices of required outdegree one. Its tail must beat exactly one of them, giving two choices. The edge between the latter pair is then forced. The two choices for the first direction and the two choices for this neighbor give exactly four tournaments.

In K_5 , fix a vertex and choose its two outgoing neighbors in $\binom{4}{2} = 6$ ways. On the remaining four vertices the required outdegrees are two at those two neighbors and one at the other two. The first assertion gives four completions for each choice, so $\text{EO}(K_5) = 6 \cdot 4 = 24$. $\square$

A balanced orientation of a graph remains balanced after subdividing an edge: the degree-two subdivision vertex has one incoming and one outgoing edge, so smoothing this directed path is a bijection. Each of L and R is obtained from K_5 by subdividing one edge. Therefore

$$\text{EO}(L) = \text{EO}(R) = 24.$$

In C , vertices 14 and 15 each have degree two. Each has two choices for its outgoing edge. The four resulting assignments to the boundary edges leave a prescribed outdegree pattern $(2, 2, 1, 1)$ on the central K_4 . By Lemma 2, each assignment has four completions, hence $\text{EO}(C) = 16$.

To justify multiplying these counts, sum outdegree minus indegree over the vertices $0, \dots, 4$ in any Eulerian orientation of W . The two edges to 14 must point in opposite directions. The analogous assertion holds for the edges joining $5, \dots, 9$ to 15. The restrictions to L and R are therefore balanced also at their subdivision vertices. Global balance at 14 and 15 then forces the restriction to C to be balanced. Conversely, any three balanced orientations of L, C, R combine to an Eulerian orientation of W . This proves

$$\text{EO}(W) = 24 \cdot 16 \cdot 24 = 9216. \tag{2}$$

Finally,

$$\frac{(4 \cdot 9216)^7}{48^{19}} = \frac{256}{243} > 1. \quad (3)$$

All quantities are positive, so taking seventh powers preserves the required comparison. No numerical approximation is used. For completeness, the exact integer cutoff for the conjectured bound at $n = 16$ is 9147, since

$$(4 \cdot 9147)^7 < 48^{19} < (4 \cdot 9148)^7.$$

3 Reduction to six branch orders

An *order- m branch* is a simple connected graph B on m vertices with two degree-three vertices a, b , called its terminals, and all other degrees four. Define its *completion* $\hat{B} = B + ab$ by adding one distinguished edge between its terminals. If ab is already present, the added edge is a separate parallel occurrence. Let

$$A_m = \max_B \text{EO}(\hat{B}),$$

where the maximum runs over all order- m branches.

Proposition 3. *Writing $M_{16} = \max_{G \in \mathcal{C}_{16}} \text{EO}(G)$, one has*

$$M_{16} = \max\{A_5 A_{10}, A_6 A_9, A_7 A_8\}. \quad (4)$$

Proof. Choose a cut vertex v of $G \in \mathcal{C}_{16}$. For a component B of $G - v$, the degree sum in G yields

$$4|V(B)| = 2|E(B)| + |\delta_G(B)|.$$

Its boundary size is positive and even. All component boundary sizes sum to four, and there are at least two components. Hence there are exactly two, each joined to v by two edges. Simplicity gives two distinct terminals in each component, so both components are branches.

A branch has at least five vertices: degree three is impossible below order four, and at order four the two nonterminals cannot have degree four. The component orders sum to 15, leaving exactly the unordered pairs $(5, 10)$, $(6, 9)$, and $(7, 8)$.

In any Eulerian orientation, summing the signed flow over a component shows that one boundary edge enters it and one leaves it. Replace the resulting directed path $a - v - b$ by the distinguished edge ab . This is a bijection to Eulerian orientations of its completion. Conversely, expand each distinguished edge into its path through v . Each branch then contributes one incoming and one outgoing edge at v , so any two completed-branch orientations combine uniquely. Thus

$$\text{EO}(G) = \text{EO}(\hat{B}_1) \text{EO}(\hat{B}_2). \quad (5)$$

There is no additional factor or boundary matching condition.

Conversely, any two branches whose orders sum to 15 can be joined to a new vertex adjacent to their four terminals, producing a graph in $\mathcal{C}_{16}$. Equation (5) proves both inequalities in (4). $\square$

Branches are allowed to have cut vertices themselves. Graphs with several possible cut vertices may occur through several decompositions; this causes only repetition and does not affect coverage. Branch isomorphisms preserve the unordered terminal pair, since these are precisely the degree-three vertices.

4 Complete generation and exact counting

4.1 An elementary complete generator

For each $m = 5, \dots, 10$, label the terminals 0, 1 and set target degrees $d_0 = d_1 = 3$ and $d_w = 4$ otherwise. Process vertices in increasing order. Before processing v , exactly the edges with an endpoint of label less than v have been fixed. Let r_w be the remaining degree needed at vertex w . Group the eligible later vertices $w > v$ with $r_w > 0$ by the pair

$$(d_w, N(w) \cap \{0, \dots, v-1\}).$$

For each group D , choose an integer k_D between zero and $|D|$. Visit every tuple with $\sum_D k_D = r_v$, and join v to the first k_D vertices of each group. Decrease the corresponding later residual degrees and set $r_v = 0$. Reject a choice if a later residual degree is outside $[0, m - v - 2]$ or the sum of later residual degrees is odd. At the terminal state, retain the graph exactly when all residual degrees vanish and the graph is connected. Restore the partial state after each recursive call.

Lemma 4. *This procedure retains at least one representative of every branch isomorphism class at each of the six orders.*

Proof. Every branch admits a labeling with terminals 0, 1. Fix a valid completion of a partial graph at step v . Vertices in each group have the same target degree and the same neighbors among all processed vertices. A permutation within a group, fixing v and all processed vertices, preserves the partial graph and target degrees. If the given completion joins v to k_D vertices of group D , such a permutation sends that neighbor set to the first k_D vertices. The resulting completion belongs to the corresponding visited tuple. None of the rejection tests can exclude a valid completion. Induction over v proves the claim. Isomorphic outputs can remain, and no deduplication is required. $\square$

4.2 Exact orientation counts

For a loopless graph or multigraph Q of even degrees, orientations are counted by the coefficient identity

$$\text{EO}(Q) = \left[\prod_{u \in V(Q)} x_u^{\deg_Q(u)/2} \right] \prod_{uv \in E(Q)} (x_u + x_v). \quad (6)$$

Selecting x_u from the edge factor chooses u as its tail. Parallel edges have separate factors.

The computation evaluates this coefficient by vertex elimination using exact integers. At vertex v , let L_v be the list of remaining edge occurrences whose other endpoint has larger label, and let t_v be the remaining outdegree demand. Choose every subset S of exactly t_v occurrences from L_v to point out of v . Each occurrence outside S , with other endpoint w , points from w to v and decreases t_w by one. Set $t_v = 0$ and continue, rejecting negative demands or demands exceeding remaining degrees. The terminal state contributes one exactly when all demands vanish. Every orientation determines exactly one sequence of choices, so the recurrence is exact. Memoization merges only identical remaining subproblems.

Two different checks were also performed. A C++ Gray-code counter exhausts all orientations after fixing the first edge direction and doubles the balanced count, using the fixed-point-free global reversal bijection. A separately implemented meet-in-the-middle counter splits the edge occurrences into two halves, enumerates their outdegree vectors with multiplicity, and pairs vectors whose sum is $(2, \dots, 2)$. Both checks reproduce every one of the 3205 representative counts, and both also count the complete 32-edge graph W directly. These checks use exact integer arithmetic and retain parallel-edge multiplicities.

4.3 Finite output and equality

The complete output is summarized in Table 1. The representative counts include duplicates. The last column likewise counts generated representatives, not isomorphism classes.

Table 1: Complete branch census with duplicates retained.

Order m	Representatives	A_m	Maximizing representatives
5	1	24	1
6	2	40	1
7	9	64	2
8	51	112	1
9	354	192	1
10	2788	384	1

As a supplementary coverage comparison, nauty 2.9.3 generation and canonicalization give 1, 2, 7, 28, 126, 719 branch isomorphism classes, respectively; the sets agree with the elementary outputs after canonicalization. The completeness proof and the decisive maxima do not depend on this comparison.

By Proposition 3, the three possible products are

$$24 \cdot 384 = 9216, \quad 40 \cdot 192 = 7680, \quad 64 \cdot 112 = 7168.$$

This proves the upper bound. Equality forces the (5, 10) decomposition with both branches maximizing. At each required order there is exactly one maximizing generated representative. Completeness therefore forces a unique branch isomorphism class at each order.

For an explicit identification, their graph6 strings are $D^{\sim}\{$ and $I\sim KGOKF@w$, with terminals 0, 1. Place the order-five branch on 0, ..., 4 and shift the order-ten branch to 5, ..., 14, then attach all four terminals to a new vertex 15. The vertex map

$$(0, 1, \dots, 15) \mapsto (5, 6, 7, 8, 9, 12, 13, 10, \\ 11, 14, 0, 1, 2, 3, 4, 15)$$

is bijective and sends its edge set exactly to the defining edge set of W . The supplementary verifier checks this edge by edge. Exchanging the two terminals of a branch introduces no different gluing, since both attach to the same new vertex. This proves uniqueness and completes the proof of Theorem 1.

5 Reproducibility and scope

The accompanying `reproducibility.zip` contains the complete elementary branch files, their exact counts, the generator and coefficient counter, the Gray-code counter, and the separately implemented meet-in-the-middle verifier and its recorded output. It also contains a read-only replay script:

```
python3 -B replay.py
```

Run this command from the extracted directory. It regenerates all six branch lists in memory, recomputes the coefficient counts, compares all 3205 records, and checks the maximizing gluing against W . It uses only the Python standard library. The accompanying README explains the additional independent checks and their software requirements. The computational upper bound and uniqueness depend on this finite enumeration and its implementation; the elementary counterexample in Section 2 does not. No proof-assistant formalization, general extremal formula, or equality classification at another order is claimed.

Computational and AI assistance. OpenAI Codex assisted with the mathematical arguments, implementation, computational verification, source comparison, and manuscript preparation. A separate model review pass reconstructed the reduction and independently implemented the meet-in-the-middle count for every representative and for W . This is computational checking, not proof-assistant certification or journal peer review. The explicit arguments and reproducible integer computations, rather than model judgments, support the stated conclusions.

References

- [1] E. Bartzos and M. Samaris, *Computing and Bounding the Number of Eulerian Orientations for Certain Classes of 4-Regular Graphs*, arXiv:2609.06701v1, 6 September 2026. <https://arxiv.org/abs/2609.06701v1>.
- [2] Z. Zeng, *Proof-Carrying Reductions for 4-Regular Girth-Nine Graphs of Excess Four*, SSRN, 2026. <https://doi.org/10.2139/ssrn.7374858>.